

Data Science

CS301

An Overview of Anova Models

Week 8

Fall 2025

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Are you here today?!



ATTENDANCE

Complete your form:

<https://forms.gle/K8yZJNJTU8soNQX27>



A New Test ...

- A statistical test to compare the *means* of multiple groups
- Allows us to determine the existence of significant differences between them.
- Commonly used in research studies to analyze the effects of different variables on an outcome (i.e., useful in scientific research fields).



Analysis of Variance (Anova)

In its simplest form, ANOVA provides a statistical test of whether two (or more) population means are equal, and therefore generalizes the t-test beyond two means.

- **Quick example:**
 - **Independent variable is social media use, and you assign groups to low, medium, and high levels of social media use.**
 - **Find out if there is a difference in hours of sleep per night**
 - **Find out more! [CLICK HERE](#)**



Independent, Dependent Variables?

- **Independent** is seemingly random or unpredictable
- **Dependent** is not random; behavior depends on independent variable.

INDEPENDENT VARIABLE

VARIABLE THAT IS CHANGED

Amount of Water



DEPENDENT VARIABLE

VARIABLE AFFECTED BY THE CHANGE

Size of Plant
Number of Leaves
Living or Dead?

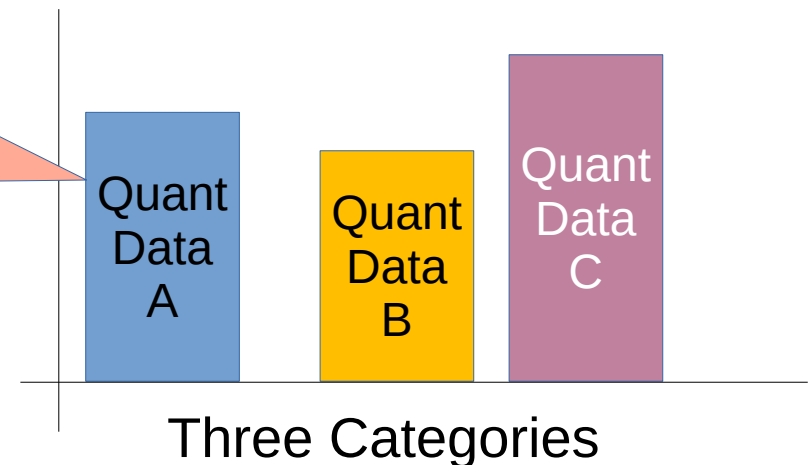




When to Use ANOVA?

- The independent variable should have at least three levels (i.e. at least three different groups or categories)
- ANOVA tells you if the *dependent* variable changes according to the level of the *independent* variable

Quantitative data is data that can be counted or measured in *numerical* values. The two main types of quantitative data are *discrete* data and *continuous* data.





ANOVA: Different from t-Test

- The Student's t-test is used to compare the means between *two* groups (levels)
- ANOVA is used to compare the means among *three or more* groups (levels)
- One way ANOVA: is a hypothesis test in which only one categorical variable or single factor is considered
- Makes comparisons between of means of three or more samples
- Each test studies differences in means and the spread of distributions (i.e., variance) across groups
- The statistical mechanisms of each test calculate statistical significance differently

Avoiding Type-1 Errors

- When there are more than two means, then we could use the **t-test** to compare the means
- But! When conducting multiple t-tests, we run into more **type-1** errors from the test
- **Using ANOVA reduces type I error rates**

- A Type I error is to reject the null hypothesis when we should have accepted it
- Erroneous conclusions that results are statistically significant when, actually, they are not significant





Did You Say, *Errors*?

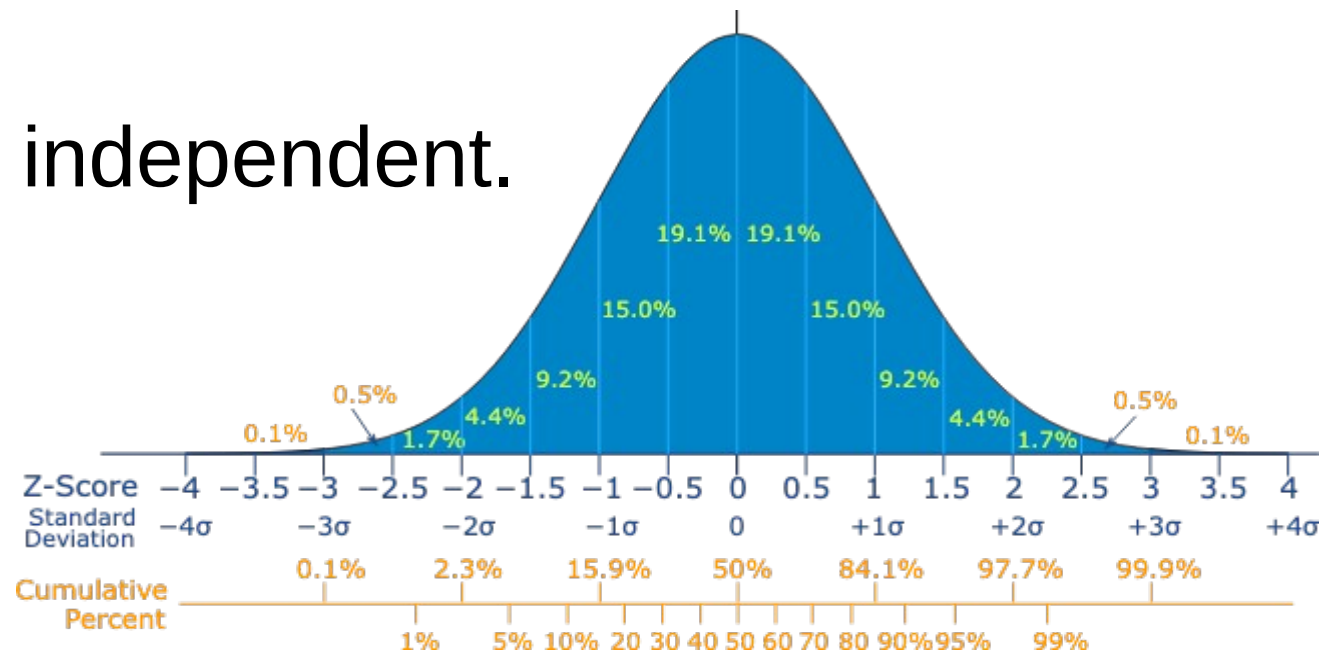
Type I and Type II Error

Null hypothesis is ...	True	False
Rejected	Type I error False positive Probability = α	Correct decision True positive Probability = $1 - \beta$
Not rejected	Correct decision True negative Probability = $1 - \alpha$	Type II error False negative Probability = β



Assumptions of ANOVA

- The responses for each factor level have a normal population distribution
- These distributions have the same variance
- The data are independent.





Testing Hypotheses in Anova

- Null hypothesis (H_0) of the ANOVA: *there is no difference between means*
 - Written as, $H_0: \mu_1 == \mu_2 == \mu_3$
- Alternative hypothesis (H_a): *the mean of at least one group is different from the others*
 - Written as, $H_a: \mu_1 != \mu_2 != \mu_3$
- For,
 - H_0 = the null hypothesis,
 - H_a = the alternative hypothesis,
 - μ_1 = the mean of population 1
 - μ_2 = the mean of population 2
 - μ_3 = the mean of population 3



Groups of Dogs

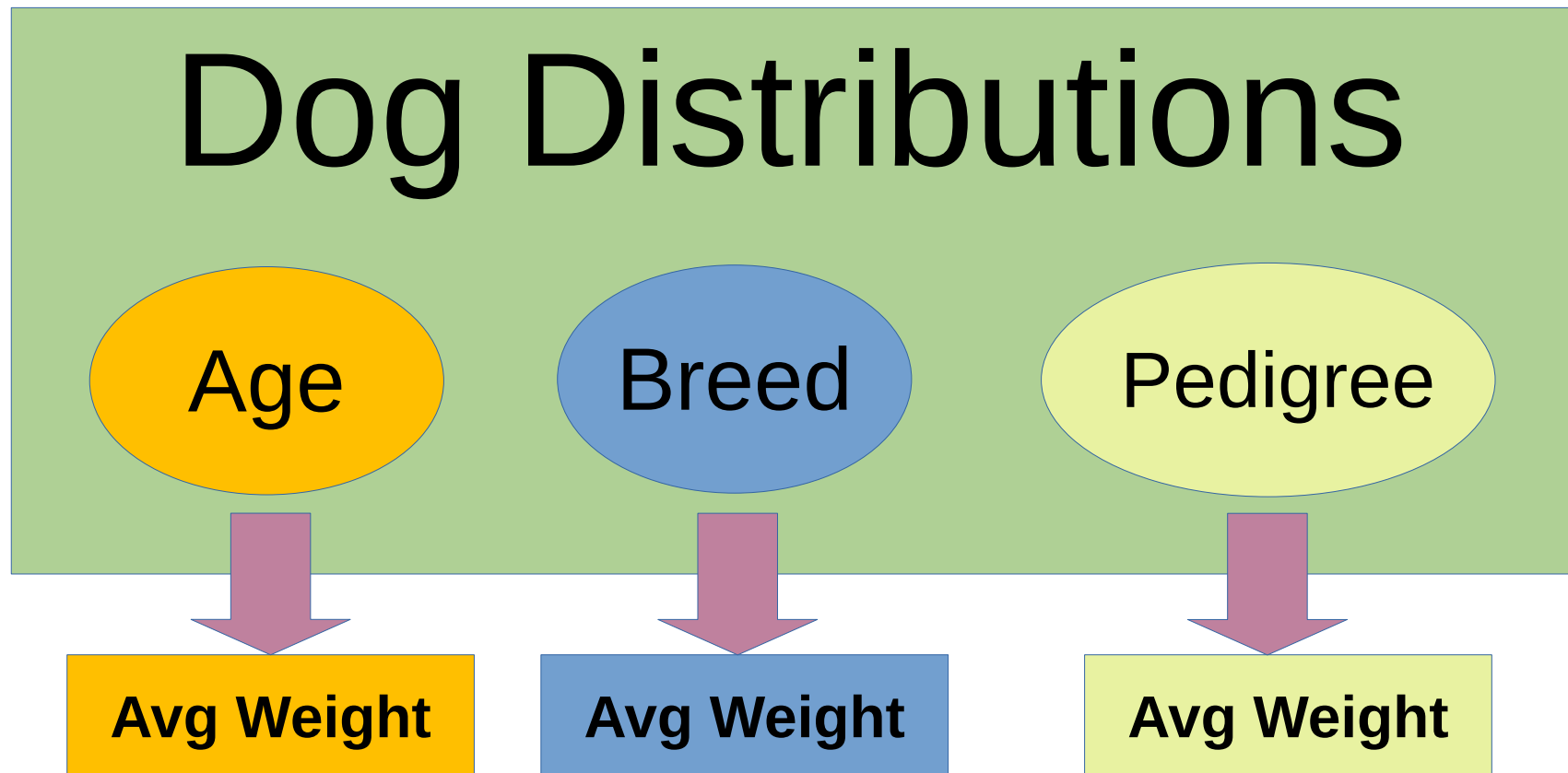


The Dog Show



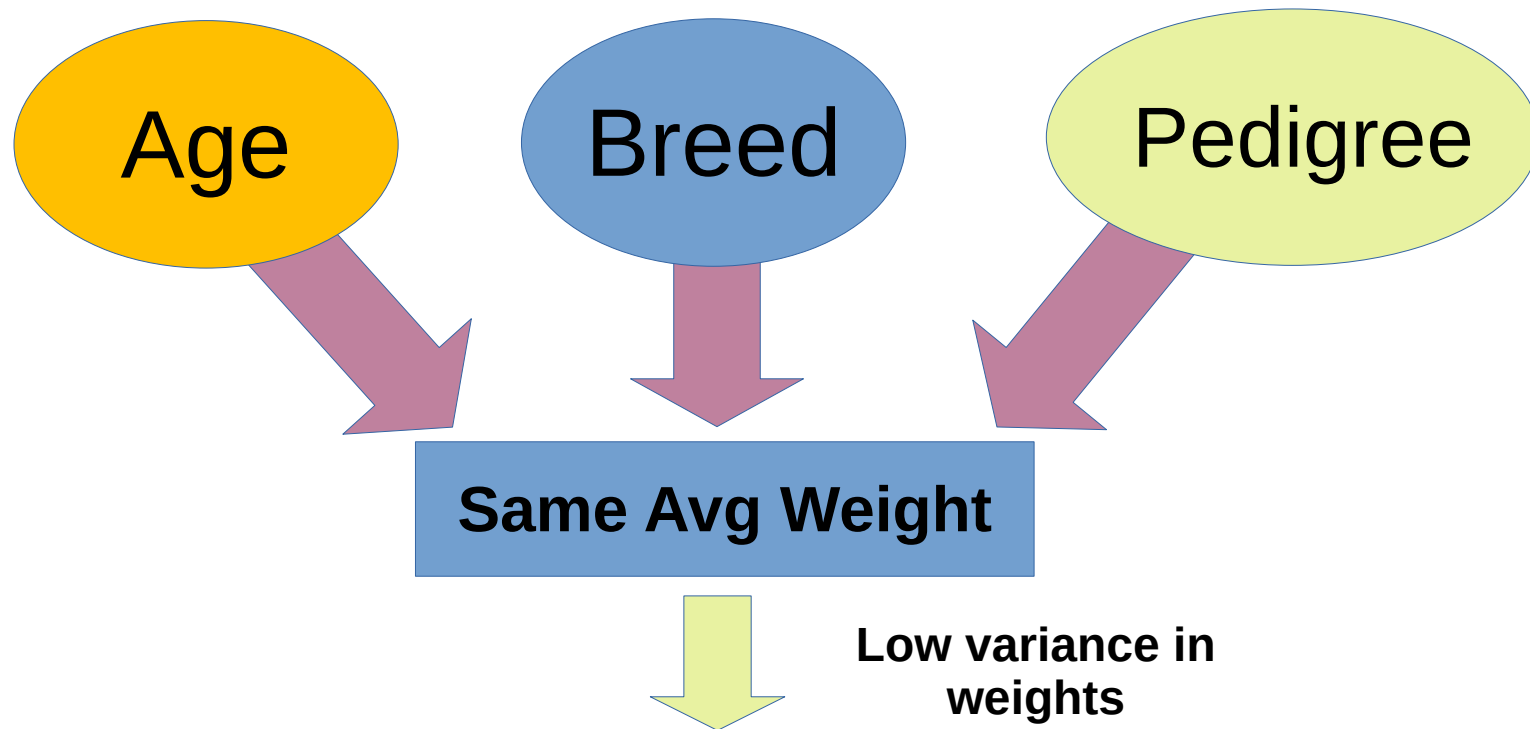
Prediction by Group Type

- We want to *predict the weight of a dog based on a certain set of characteristics* for each dog (age, breed, and pedigree)





Explain Dog Weights According to Groups



***If groups have the same mean,
then it isn't reasonable to conclude
that the groups are, in fact,
separate in any meaningful way***

Similar Means From Low Variances

- Separate dogs into distributions having low variance of dog weights (a *heterogeneous* group) ...
- The weight-means between groups should be distinct
- If weight-means were similar between groups, then the groups would be similar.

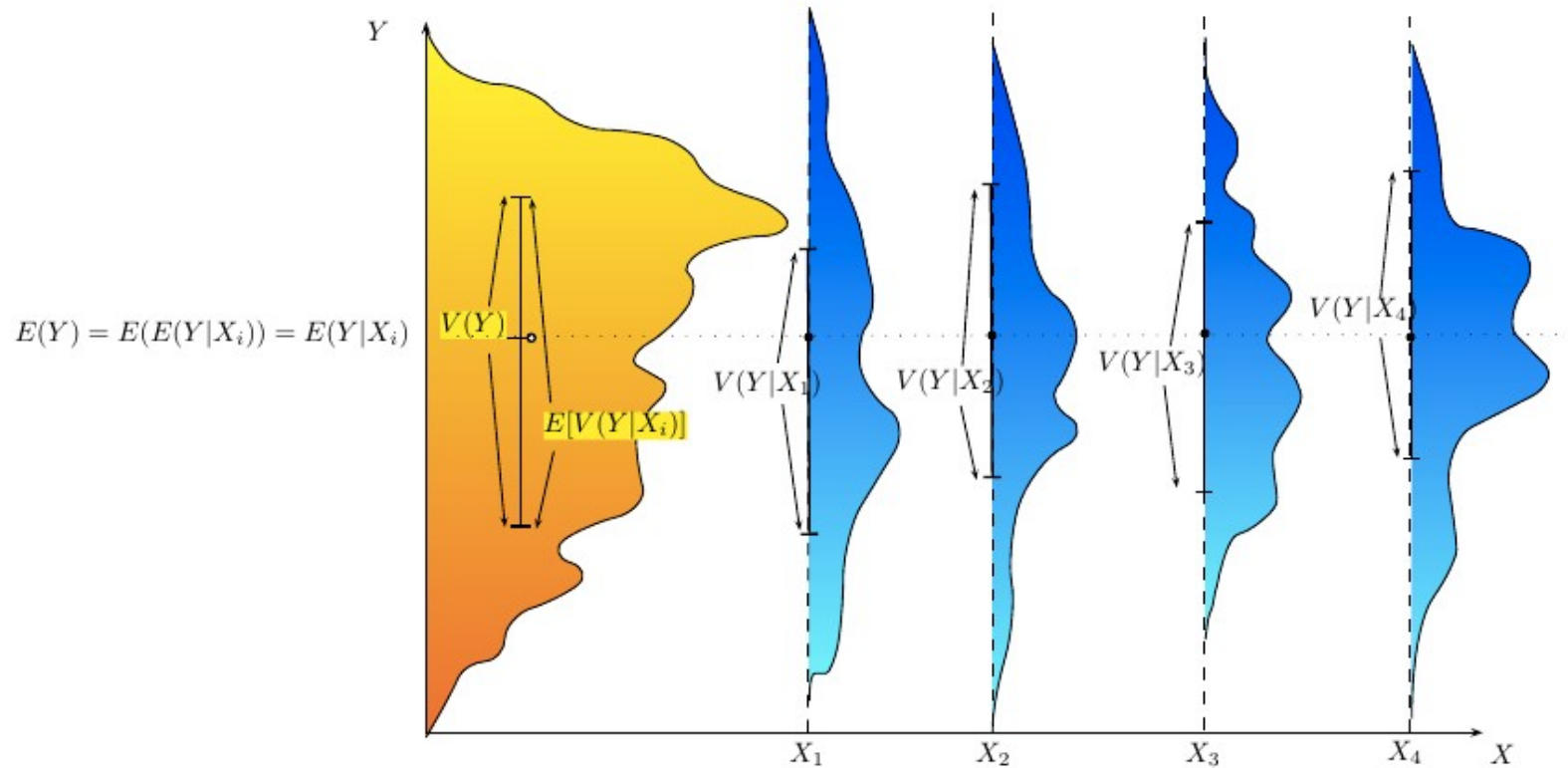
We create two groups:
are-cute and ***have floppy ears***.

The mean-weights would
be similar from each group.

All these Beagles are (both)
cute and
have ***floppy ears***!

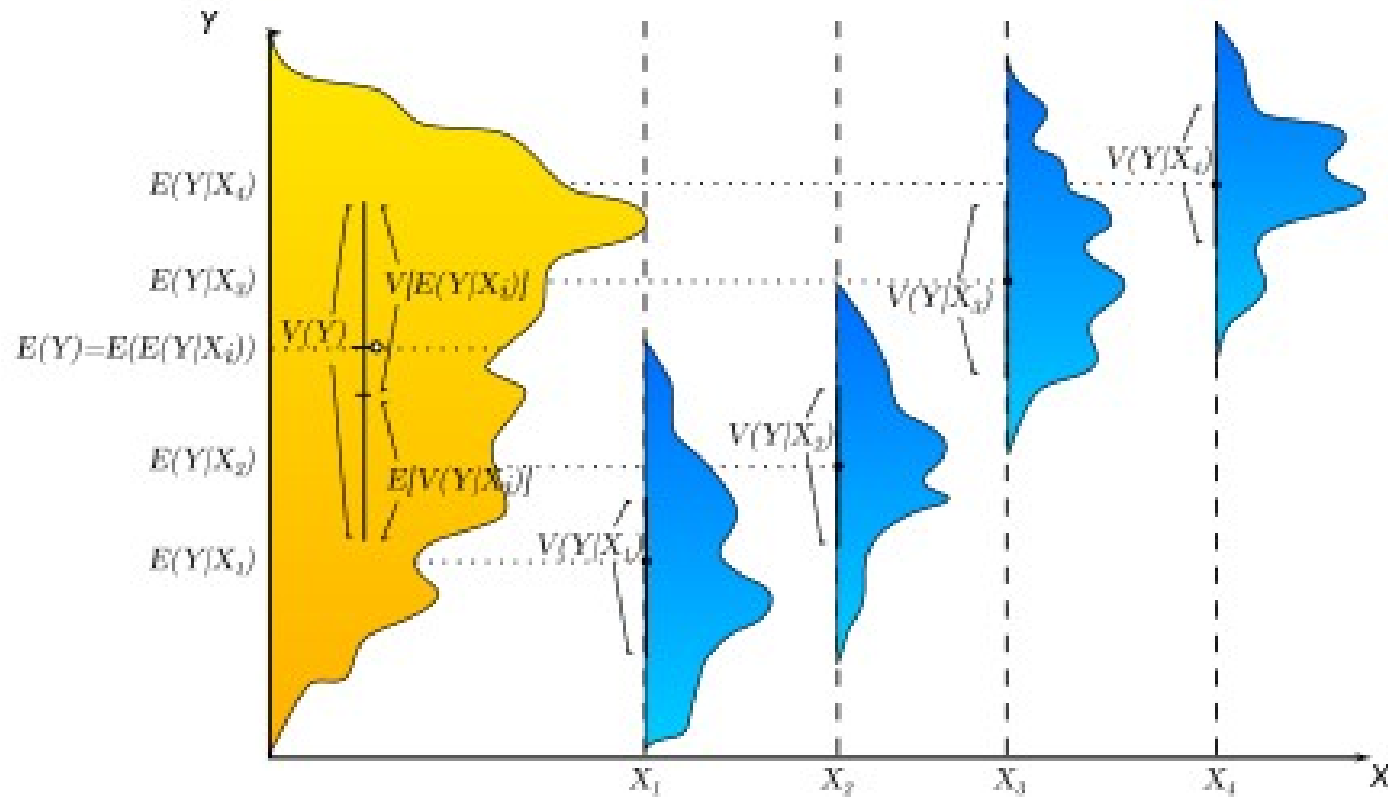


Comparison of Variances No Fit



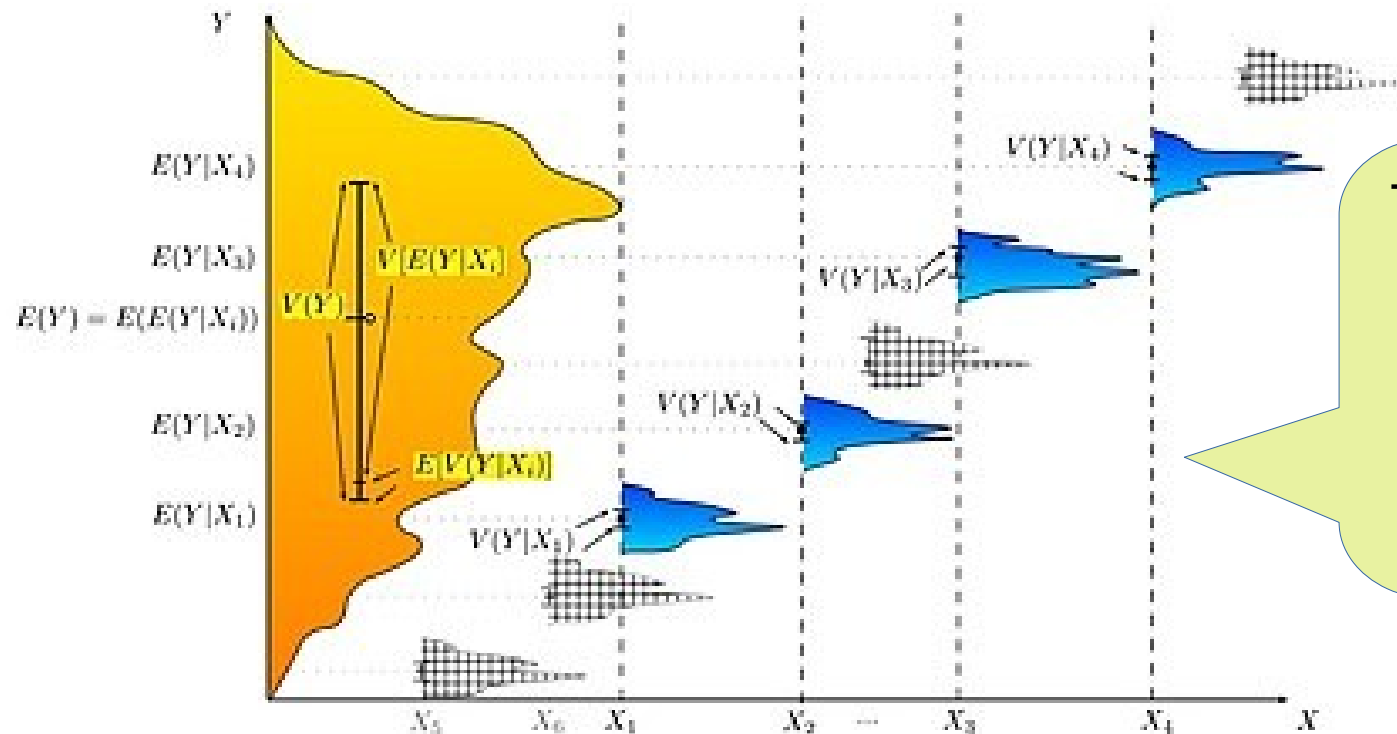
- Orange: mean-weights of all dogs
- Blues: All mean-weights by types of groups
- *Groups do not explain variation in distributions*

Comparison of Variances Better/Fair Fit



- Orange: mean-weights of all dogs
- Blues: All mean-weights by types of groups
- Groups are **becoming distinguishable** by mean-weights

Comparison of Variances Better/Fair Fit



The heaviest dogs are likely to be bigger breeds, while the lighter breeds tend to be smaller dogs.

- Orange: mean-weights of all dogs
- Blues: All mean-weights by types of groups
- ***Groups are distinguishable by mean-weights***



Group-making: Lighter Dogs



Chihuahua



Pomeranian



Shih Tzu



Yorkshire Terrier



Pug



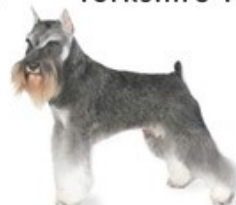
Dachshund



Miniature Pinscher



Bichon Frise



Miniature Schnauzer



Toy Poodle



Chiweenie



English Bulldog



Pekingese



Bolonka



Maltese



Phalene



Basset Hound



Papillon



Havanese



Löwchen



Cavalier King
Charles Spaniel



French Bulldog

Dog groups to similar types of weight

Group-making: Heavy Dogs



Dog groups to similar types of weight



So What Then?

ANOVA provides a statistical test of equality of weight-means across the groups of dogs.

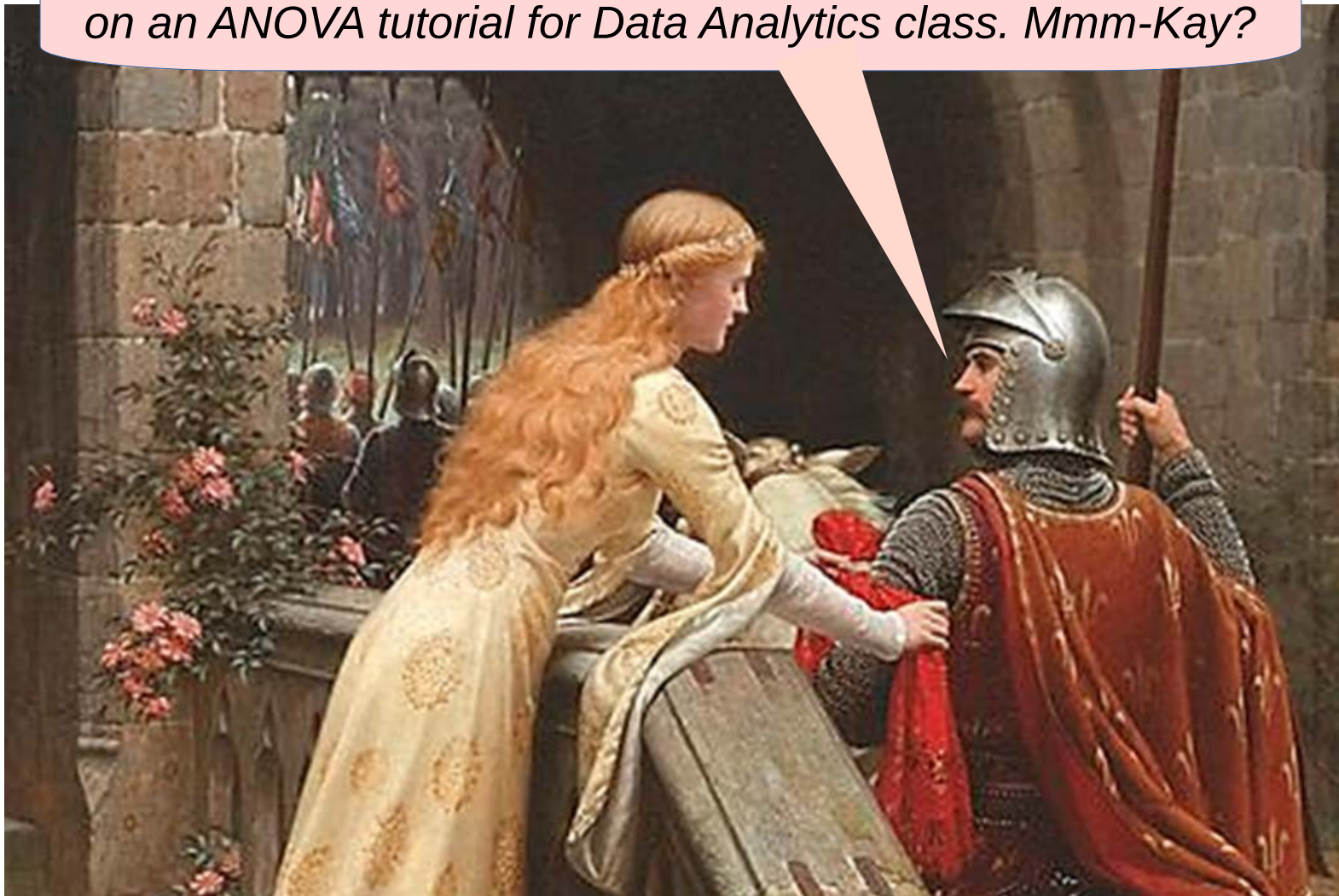
**An ANOVA
generalizes
the t-test beyond
two means.**





Let's Hit The Code

That sounds like fun, but first I'm just gonna work on an ANOVA tutorial for Data Analytics class. Mmm-Kay?



Follow the code from the Tutorial and address some questions. URL on next slide.



Let's Look at Some Code!

```
# clear out the variables
rm(list = ls())
# clear out all plots from previous
work.
graphics.off()
# clear the console
cat("\014")

# install libraries
if(!require('tidyverse')) {
  install.packages('tidyverse')
  library('tidyverse')
}
```

**Clean that
work space!**





Create Random Data to Test

```
# Set seed for reproducibility  
set.seed(123)
```

```
# Generate synthetic data  
group1 <- rnorm(30, mean = 5, sd = 1)  
group2 <- rnorm(30, mean = 7, sd = 1)  
group3 <- rnorm(30, mean = 6, sd = 1)
```

Randomly generated and normal distribution

```
> group1  
[1] 4.439524 4.769823 6.558708 5.070508 5.129288 6.715065 5.460916 3.734939 4.313147  
[10] 4.554338 6.224082 5.359814 5.400771 5.110683 4.444159 6.786913 5.497850 3.033383  
[19] 5.701356 4.527209 3.932176 4.782025 3.973996 4.271109 4.374961 3.313307 5.837787  
[28] 5.153373 3.861863 6.253815
```



Groups Into Data Frame

```
# Set seed for reproducibility
set.seed(123)

# Generate synthetic data
group1 <- rnorm(30, mean = 5, sd = 1)
group2 <- rnorm(30, mean = 7, sd = 1)
group3 <- rnorm(30, mean = 6, sd = 1)

# Place into a data frame
data <- data.frame(
  value = c(group1, group2, group3),
  group = factor(rep(c("Group 1", "Group 2", "Group 3"),
    each = 30))
)
```

Run the Test, Visualize to Understand Results

```
# Perform ANOVA
anova_result <- aov(value ~ group, data = data)

# Display the summary of the ANOVA
summary(anova_result)

# Visualize means by boxplot
ggplot(data, aes(x = group, y = value)) +
  geom_boxplot(fill = c("lightblue", "lightgreen",
"lightcoral")) +
  theme_minimal() +
  labs(title = "Boxplot of Values by Group",
       x = "Group",
       y = "Value")
```



Check the Significance

The p -value is significant

```
> # Display the summary of the ANOVA  
> summary(anova_result)
```

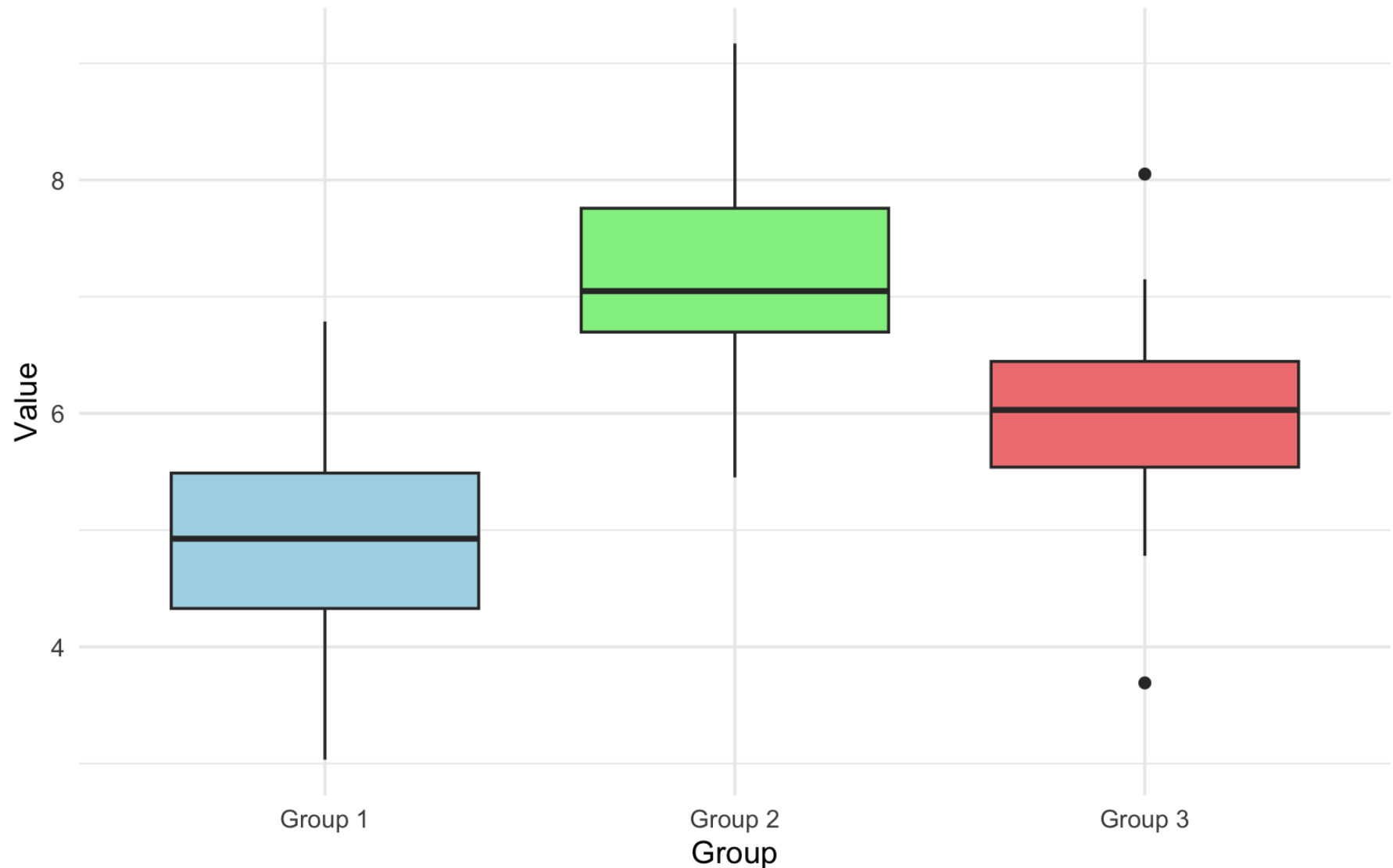
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
group	2	74.32	37.16	46.14	2.19e-14 ***
Residuals	87	70.08	0.81		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1



Visualize Difference Between Group Means

Boxplot of Values by Group





Anyone For Completing a Tutorial?

ANOVA in R | A Complete Step-by-Step Guide with Examples

Published on March 6, 2020 by [Rebecca Bevans](#). Revised on November 17, 2022.

ANOVA is a [statistical test](#) for estimating how a quantitative [dependent variable](#) changes according to the levels of one or more categorical [independent variables](#). ANOVA tests whether there is a difference in means of the groups at each level of the independent variable.

Click here!

<https://www.scribbr.com/statistics/anova-in-r/>